MA3K7 Final Project Rubric

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1 Entry

I know

From the problem statement, I know a few things straight away.

- there are three species of particles, namely A, B, C.
- the particles stay inside the box, so the total number of particles is preserved.
- if two different species collide, they **both** turn into the third species, so

$$A + B \to 2C, \qquad B + C \to 2A, \qquad C + A \to 2B.$$

Notation

For brevity, I use the following

- Let (a, b, c) denote the number of particles of species A, B, C respectively.
- Let $N = a + b + c$ be the total number of particles, which stays fixed throughout.

I want

What I want, is an idea of what repeated collisions actually do. For instance, if I keep applying collisions,

- does the system eventually become a single species? If so, does the initial number of each particle affect this?
- can the system keep cycling forever?
- if the outcome depends on the starting condition, what determines it?
- are there any properties that remain unchanged under every collision? If so, can they be used to classify all possible outcomes?

This resembles systems where an invariant determines the long-term behaviour, so I expect something like that to exist here.

Introduce

Because the choice of collisions is non-deterministic, it is not immediately clear what typical trajectories look like. My first instinct was therefore not to try to prove anything straight away, but instead to simulate the process using a Monte Carlo approach in Python and see how the initial state evolves over time.

Before doing that, I first computed a few small cases by hand to look for patterns and get a better idea of how the system behaves.

Small cases by hand

I first define what I mean by trivial cases.

Define

Definition 1.1. *A configuration is called a pure (or trivial) state if all particles are of the same species, for example* $(N, 0, 0)$*, and similarly for the other species.*

In this case, no further collisions can change the state, so the system remains static.

From now on, I will focus on non-trivial systems where at least two species are present, since only in these cases can interactions occur and the system can actually evolve.

Case $N = 2$

Up to relabelling, I will treat $(1, 1, 0)$ as the representative case, since $(0, 1, 1)$ and $(1, 0, 1)$ behave in the same way with the species permuted. The only non-trivial configuration is

$$(1, 1, 0).$$

A single collision gives

$$(1, 1, 0) \to (0, 0, 2),$$

so the system immediately collapses to a single species, namely the one that was not present initially.

Case $N = 3$

There are now more possible configurations.

Starting from

$$(2, 1, 0),$$

one collision gives

$$(2, 1, 0) \to (1, 0, 2),$$

and continuing,

$$(1, 0, 2) \to (0, 2, 1) \to (2, 1, 0).$$

Aha

So instead of collapsing, the system cycles through three states and returns to where it started. This suggests that the system does not always collapse to a single species and that there may be an underlying structure governing the evolution. In particular, the cyclic nature of the transitions hints at a possible modular structure behind the system.

On the other hand, starting from

$$(1, 1, 1),$$

any collision produces two particles of the third type, for example

$$(1, 1, 1) \to (0, 0, 3).$$

Observation

So in this case, some configurations collapse immediately, while others do not. This suggests that the behaviour of the system depends on the initial configuration of the particles, and may be determined entirely by it.

Conjecture 1. *The long-term behaviour of the system is determined entirely by the initial configuration.*

Case $N = 4$

I first considered

$$(3, 1, 0).$$

One possible sequence is

$$(3, 1, 0) \to (2, 0, 2) \to (1, 2, 1).$$

From here, the next collision can either give

$$(1, 2, 1) \to (0, 4, 0)$$

or return to

$$(1, 2, 1) \to (3, 1, 0).$$

So from this configuration, the system can either collapse to a single species or return to a previous state.

I then looked at

$$(2, 1, 1).$$

One possible sequence is

$$(2, 1, 1) \to (4, 0, 0)$$

directly, or

$$(2, 1, 1) \to (1, 0, 3) \to (0, 2, 2) \to (2, 1, 1),$$

which forms a cycle.

NOTE

Notably, this overlaps with the earlier trajectory from $(3, 1, 0)$. In particular, the state $(1, 2, 1)$ has the same shape, up to relabelling of the species, as states appearing in that trajectory. This suggests that these configurations are not isolated, but belong to a connected set of states that the system can move between.

So for $N = 4$, both behaviours appear:

- some configurations can collapse to a single species,
- others can cycle and return to their starting point.

INTUITION

At this point, even when the system returns to a previous configuration, it still seems possible for it to move towards a pure state, as in the $(3, 1, 0)$ example.

In particular, although the system can revisit earlier states, there are still sequences of collisions that lead to a pure state. So at least in the examples I had looked at, returning to a previous state did not rule out the possibility of eventually reaching a pure state.

RE-VISIT

However, I do not yet know which initial configurations actually admit such a path. The examples suggest that some do and some do not, but I do not yet have a way to distinguish them.

TRY

My next thought was to look for something that stays unchanged. The most obvious candidates were the simple differences

$$a - b, \qquad b - c, \qquad c - a.$$

For instance, starting from

$$(1, 1, 0),$$

I get

$$a - b = 0, \qquad b - c = 1, \qquad c - a = -1.$$

After one collision,

$$(1, 1, 0) \to (0, 0, 2),$$

so now

$$a - b = 0, \qquad b - c = -2, \qquad c - a = 2.$$

BUT

So these are not preserved. this was still useful, because it suggested that any invariant is unlikely to be a simple linear one.

What kept coming back was the cyclic nature of the rule:

$$A + B \to C, \qquad B + C \to A, \qquad C + A \to B.$$

It does not behave like an additive system, but it does look as though the species are arranged in a cycle.

CONJECTURE

That made me think that instead of working over the integers directly, it might be more natural to look at the system modulo 3.

Conjecture 2. *There exists a property that is preserved under every collision, which depends only on* (a, b, c) *modulo* 3.

At this point I did not really have a proof for this conjecture. It was more of a feeling that something like the differences between the particle counts might matter modulo 3, rather than exactly.

INTRODUCE

Before pushing that idea further, I decided to check what actually happens in practice. As mentioned earlier, I wrote a short Python script which generates random trajectories starting from a given state (a, b, c).

```
# code in github / Final_Project/src/Monte_Carlo.py
```

Rather than looking at just one starting point, I fixed $N = 6$ and ran the simulation over all non-trivial configurations with $a + b + c = N$. For each configuration, I repeated the simulation many times and recorded whether the trajectory collapsed to a single species, or returned to a state that had already appeared earlier in the same run.

```
(0, 1, 5) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(0, 2, 4) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(0, 3, 3) counts = {'collapsed': 59, 'repeat': 441, 'stuck': 0}
(0, 4, 2) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(0, 5, 1) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(1, 0, 5) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(1, 1, 4) counts = {'collapsed': 95, 'repeat': 405, 'stuck': 0}
(1, 2, 3) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(1, 3, 2) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(1, 4, 1) counts = {'collapsed': 70, 'repeat': 430, 'stuck': 0}
(1, 5, 0) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(2, 0, 4) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(2, 1, 3) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(2, 2, 2) counts = {'collapsed': 51, 'repeat': 449, 'stuck': 0}
(2, 3, 1) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(2, 4, 0) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(3, 0, 3) counts = {'collapsed': 51, 'repeat': 449, 'stuck': 0}
(3, 1, 2) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(3, 2, 1) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(3, 3, 0) counts = {'collapsed': 56, 'repeat': 444, 'stuck': 0}
(4, 0, 2) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(4, 1, 1) counts = {'collapsed': 84, 'repeat': 416, 'stuck': 0}
(4, 2, 0) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(5, 0, 1) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
(5, 1, 0) counts = {'collapsed': 0, 'repeat': 500, 'stuck': 0}
```

Figure 1: Simulation results for all non-trivial states with $N = 6$

OBSERVATION

Looking more closely at the output, the configurations appeared to split into two types:

1. states for which the simulation frequently reaches a single-species configuration,
2. states for which no such collapse was ever observed, and the trajectory always returned to previously visited states.

This strongly suggests that the behaviour is not probabilistic in nature, but instead determined by some underlying structural property of the initial state.

STUCK

This was also the point where I got stuck a little bit. I had a consistent pattern in the data, but I still could not see a clear reason for why one configuration leads to collapse while another only produces repeated states.

The total number of particles clearly does not help, since it is fixed, and the obvious quantities like $a - b$, $b - c$, $c - a$ failed. So I thought the underlying structure must be something less direct.

RE-ENTER

So instead of continuing to guess blindly, I tried to narrow the question down. Rather than asking what happens in general, I focused on whether there exists some quantity that survives every collision, and if so, whether it completely determines which states are reachable.

2 Attack

NOTE At this point, I wanted to stop looking at full trajectories and instead understand what a single collision actually does to the state. So I wrote out the updates explicitly.

From the rules,

$$\begin{aligned} A + B \to 2C \quad &\Rightarrow \quad (a, b, c) \to (a - 1, b - 1, c + 2), \\ B + C \to 2A \quad &\Rightarrow \quad (a, b, c) \to (a + 2, b - 1, c - 1), \\ C + A \to 2B \quad &\Rightarrow \quad (a, b, c) \to (a - 1, b + 2, c - 1). \end{aligned}$$

TRY Earlier I had already tried the simple differences $a - b$, $b - c$, $c - a$, and saw that they are not preserved. However, looking at how they fail turns out to be more informative.

Take for example $A + B \to 2C$:

$$(a, b, c) \to (a - 1, b - 1, c + 2).$$

Then

$$\begin{aligned} (a - b) &\to (a - 1) - (b - 1) = a - b, \\ (b - c) &\to (b - 1) - (c + 2) = (b - c) - 3, \\ (c - a) &\to (c + 2) - (a - 1) = (c - a) + 3. \end{aligned}$$

OBSERVATION So the differences are not fixed, but they only change by multiples of 3.

At first this just looked like a coincidence, but the same behaviour appears for the other two collisions.

For $B + C \to 2A$,

$$(a, b, c) \to (a + 2, b - 1, c - 1),$$

which gives

$$(a - b) \to (a - b) + 3, \quad (b - c) \to b - c, \quad (c - a) \to (c - a) - 3.$$

For $C + A \to 2B$,

$$(a, b, c) \to (a - 1, b + 2, c - 1),$$

so

$$(a - b) \to (a - b) - 3, \quad (b - c) \to (b - c) + 3, \quad (c - a) \to c - a.$$

AHA So in every case, each of $a - b$, $b - c$, $c - a$ either stays the same or shifts by ± 3. That means their values modulo 3 never change.

Proposition 2.1. *The quantities* $(a - b) \bmod 3$, $(b - c) \bmod 3$, *and* $(c - a) \bmod 3$ *are invariant under every collision.*

Proof. Each collision updates (a, b, c) by adding one of the vectors

$$(-1, -1, +2), \quad (+2, -1, -1), \quad (-1, +2, -1).$$

In each case, the change in $(a - b)$, $(b - c)$, $(c - a)$ is given by

$$(0, -3, +3), \quad (+3, 0, -3), \quad (-3, +3, 0),$$

respectively.

So every difference either stays the same or changes by a multiple of 3. Hence their values modulo 3 are preserved under every collision. □

Re-visit

This also confirms the conjecture from the Entry section, where I suspected that working modulo 3 might be natural. What initially appeared as a guess is now forced by the structure of the updates.

Check

I also verified this computationally by tracking the quantities

$$((a-b),(b-c),(c-a)) \bmod 3$$

along simulated trajectories. In all cases tested, these values remained constant, consistent with the proposition.

```
# code in github / Final_Project/src/Invariance.py
```

```
(1, 1, 4) (0, 0, 0) True
(2, 2, 2) (0, 0, 0) True
(3, 1, 2) (2, 2, 2) True
(0, 1, 5) (2, 2, 2) True
```

Figure 2: Verification of invariance modulo 3 along simulated trajectories

Figure 2 illustrates that for each initial configuration, the quantities remain unchanged along the trajectory, consistent with the proposition above.

Conjecture

From the simulation results in the Entry section, I was led to the following

Conjecture 3. *Suppose* $N \equiv 0 \pmod 3$. *Then the system can reach a single-species state if and only if*

$$a \equiv b \equiv c \pmod 3.$$

This pattern was consistently observed in the simulations for $N = 6$, where all configurations that collapsed satisfied $a \equiv b \equiv c \pmod 3$.

Note

The assumption $N \equiv 0 \pmod 3$ is important here. The examples with $N = 4$ already show that without this restriction, the condition $a \equiv b \equiv c \pmod 3$ is too strong. For instance, $(3, 1, 0)$ can still reach a pure state.

Analyse

To understand the conjecture better, I first try to prove one direction. In particular, I want to see why this condition should be necessary.

For example, in a pure state such as $(N, 0, 0)$, the differences are

$$a - b = N, \qquad b - c = 0, \qquad c - a = -N.$$

So modulo 3 they become

$$(a-b, b-c, c-a) \equiv (N, 0, -N) \pmod 3.$$

Since these quantities are invariant under every collision, any initial configuration that reaches a pure state must lie in the same invariant class as some pure state.

Now if $N \equiv 0 \pmod 3$, then in every pure state these three differences are all congruent to $0 \pmod 3$. So in that case, any configuration that reaches a pure state must satisfy

$$a - b \equiv b - c \equiv c - a \equiv 0 \pmod 3,$$

which is equivalent to

$$a \equiv b \equiv c \pmod 3.$$

Aha

So under the assumption $N \equiv 0 \pmod 3$, any configuration that reaches a pure state must

satisfy

$$a \equiv b \equiv c \pmod 3.$$

This shows that the condition is necessary.

CHECK To test the strength of this condition, I also looked for possible counterexamples. In particular, I checked whether configurations with $N \equiv 0 \pmod 3$ but $a \not\equiv b \equiv c \pmod 3$ could still collapse. However, in all small cases and simulations, no such examples appeared.

STUCK At this point I had an invariant and a strong empirical pattern, but I was not yet sure whether this fully determines which states can reach a pure configuration.

RE-ENTER To make further progress, I decided to separate the problem into two parts. First, to understand precisely what the invariant forbids. Second, to determine whether the condition $a \equiv b \equiv c \pmod 3$ is also sufficient when $N \equiv 0 \pmod 3$.

3 Solution

REMIND From the Attack section, we know that the quantities

$$(a-b) \bmod 3, \quad (b-c) \bmod 3, \quad (c-a) \bmod 3$$

are invariant under every collision.

Under the assumption

$$N \equiv 0 \pmod 3,$$

any pure state has all three of these differences congruent to $0 \pmod 3$. So if a configuration can reach a pure state, it must satisfy

$$a \equiv b \equiv c \pmod 3.$$

It remains to show that this condition is also sufficient.

Part 1: Reducing to two species

SETTING Suppose for some $x \in \{0, 1, 2\}$

$$a \equiv b \equiv c \equiv x \pmod 3.$$

If all three of a, b, c are positive, we may apply

$$B + C \to 2A,$$

which gives

$$(a, b, c) \to (a+2, b-1, c-1).$$

This reduces both b and c by 1, while preserving the invariant.

Repeating this $B+C$ operation as many times ($\min(b, c)$ times), let $p = \min(b, c)$. We then reach

$$(a + 2p,\ b - p,\ c - p),$$

so that the smaller of b, c is reduced to zero. Hence we obtain a configuration of the form

$$(a', b', 0),$$

or symmetrically $(a', 0, c')$.

CHECK At this point, I want to understand what happens to the congruence condition under this reduction.

Suppose for concreteness that $c \leq b$, so that

$$a' = a + 2c, \qquad b' = b - c.$$

Since

$$a \equiv b \equiv c \equiv x \pmod 3,$$

we get

$$b' \equiv b - c \equiv x - x \equiv 0 \pmod 3,$$

and

$$a' \equiv a + 2c \equiv x + 2x = 3x \equiv 0 \pmod 3.$$

since $3x$ is a multiple of 3.

So in this case,

$$a' \equiv b' \equiv 0 \pmod 3.$$

The case $(a', 0, c')$ is symmetric.

CONCLUSION This shows that any configuration satisfying

$$a \equiv b \equiv c \pmod 3$$

can be reduced to a configuration of the form

$$(a', b', 0)$$

with

$$a' \equiv b' \equiv 0 \pmod 3.$$

So from this point on, it suffices to understand what happens in the two-species case.

Part 2: Reaching a pure state from two species

CONTINUATION From Part 1, we may assume that the system is in a configuration of the form

$$(a, b, 0)$$

with

$$a \equiv b \equiv 0 \pmod 3.$$

So we may write

$$b = 3m$$

for some integer $m \geq 0$.

If $m = 0$, then $b = 0$ and $(a, 0, 0)$ is already a pure state.

Instead, if $m \neq 0$, consider the following sequence of three collisions

1. $(a, b, 0) \xrightarrow{A+B \to 2C} (a-1, b-1, 2)$
2. $(a-1, b-1, 2) \xrightarrow{B+C \to 2A} (a+1, b-2, 1)$
3. $(a+1, b-2, 1) \xrightarrow{B+C \to 2A} (a+3, b-3, 0)$

The net effect is

$$(a, b, 0) \longrightarrow (a + 3, b - 3, 0).$$

When $m \geq 1$, we have $b \geq 3$, so each of the three collisions above is valid and all intermediate states have non-negative coordinates.

AHA

Each application reduces b by 3 and increases a by 3. Repeating this m times gives

$$(a, b, 0) \to (a + 3m, 0, 0) = (N, 0, 0).$$

Conclusion

Proposition 3.1. *Suppose* $N \equiv 0 \pmod 3$. *If*

$$a \equiv b \equiv c \pmod 3,$$

then the system can reach a pure state.

Proof. Part 1 reduces the problem to the two-species case, and Part 2 shows that any such configuration can be transformed into a pure state in finitely many steps. □

COMBINE

Combining this with the necessary condition from the Attack section, we obtain the main result.

Theorem 1. *Suppose* $N \equiv 0 \pmod 3$. *Then the system can reach a single-species state if and only if*

$$a \equiv b \equiv c \pmod 3.$$

NECESSITY

Proof. This was established in the Attack section using the invariant

$$(a - b) \bmod 3, \quad (b - c) \bmod 3, \quad (c - a) \bmod 3.$$

Under the assumption $N \equiv 0 \pmod 3$, every pure state has these three quantities equal to 0 (mod 3), so any configuration that reaches a pure state must satisfy

$$a \equiv b \equiv c \pmod 3.$$

SUFFICIENCY

This follows from Proposition 3.1. Part 1 reduces the system to a two-species configuration, and Part 2 constructs an explicit finite sequence of collisions leading to a pure state. □

4 Review and Extensions

RE-VISIT

Looking back at the original questions in the Entry section, the picture is much clearer now.

- the system does not behave randomly; whether a pure state can be reached is determined by the initial configuration,
- there is an invariant, namely the pairwise differences modulo 3,
- under the assumption $N \equiv 0 \pmod 3$, this invariant completely determines whether collapse is possible,
- and if the condition fails, the system cannot reach a pure state and instead remains trapped among mixed configurations.

So the behaviour that initially looked complicated turns out to be governed by a simple modular structure.

REVIEW

The key step in the solution was to stop thinking about long trajectories and instead ask what a single collision actually changes. Once I wrote the updates out explicitly, the structure modulo 3 appeared quite naturally.

One thing that became clearer only later is that the invariant is not something artificial added afterwards, but is already built into the rule itself. The system

$$A + B \to C, \qquad B + C \to A, \qquad C + A \to B$$

is cyclic from the start, and the fact that the differences only change by multiples of 3 reflects this symmetry.

INSIGHT

At the beginning, I tried simple quantities such as $a - b$, $b - c$, and $c - a$, but these were not preserved exactly. What mattered was not their exact values, but their residue classes modulo 3. In that sense, the right invariant was already there, but I only saw it after changing viewpoint.

Possible extensions

EXTENSION

The first extension that now feels natural is to ask what happens to states that cannot collapse. In the main result, I classified when a pure state is reachable, but this leaves open the structure of the remaining states.

For example, when $N = 3$, the state

$$(2, 1, 0) \to (1, 0, 2) \to (0, 2, 1) \to (2, 1, 0)$$

lies in a cycle of length 3. This suggests looking more systematically at the non-collapsing part of the system.

A useful way to think about this is as a directed graph: each state (a, b, c) is a vertex, and each legal collision gives a directed edge to a new state. In that language, the problem becomes more structural. A state can collapse exactly when there is a directed path from that vertex to a pure state. States that cannot collapse must lie in components with no outgoing path to a pure vertex.

This seems like a good extension because it takes the main theorem and reinterprets it in a different language, rather than just repeating the same argument. It would also fit naturally with Python, since for small fixed N one could generate the whole state graph and inspect its connected structure directly.

TRY

A second extension is to change the collision rule itself. For instance, instead of

$$A + B \to 2C,$$

one could study the more general rule

$$A + B \to nC,$$

and similarly for the other pairs.

In that case, the total number of particles is no longer preserved, so the original argument cannot be used unchanged. However, the same calculation suggests a related modular structure. Under

$$(a, b, c) \to (a - 1, b - 1, c + n),$$

the differences change by multiples of $n+1$, so it is natural to suspect that an invariant modulo $n+1$ may appear.

What is less clear is whether the classification is still as clean as in the original problem. Since the total number of particles is no longer fixed, the state space itself changes, so even formulating the right analogue of collapse becomes less straightforward. That is exactly what makes the extension interesting: the same modular idea seems to survive, but the global structure may be quite different.

Extension

Another possible direction is to increase the number of species. With four species arranged cyclically, one could try rules such as

$$A + B \to C, \qquad B + C \to D, \qquad C + D \to A, \qquad D + A \to B.$$

The original problem suggests that the appearance of modulo 3 is not accidental, but tied to the cyclic structure of three species. So it is natural to ask whether a similar invariant appears for larger cyclic systems, or whether the three-species case is unusually rigid.

Reflection

Finally, one thing I have not really understood yet is the detailed structure of the cycles themselves. The theorem tells me exactly when collapse is possible, but it says much less about the geometry of the states that do not collapse. Questions such as

- how many distinct cycles exist for a fixed N,
- how long these cycles can be,
- and how the cycle structure depends on the invariant class

would all be natural next steps.

So although the original question can be answered quite cleanly, the system still seems to contain more structure than just the collapse criterion. In that sense, the solution feels like an ending to the first question rather than the end of the problem itself.

Code

The code for this assignment can be found on my GitHub page:
`https://github.com/cathal0317/Problem_Solving_RUBRIC`